



What is HTTP/2?

A new version of the HTTP protocol was published as a draft by the IETFForce, soon to replace it. <http://dgit.in/HTTP2Prot>

Recyclable 3D filament

Dimension Polymers is trying to develop recycled plastic filaments that can be used in 3D printing <http://dgit.in/3DPrntFila>

Streaming Devices

range. Also, high-powered SoCs can run without being underclocked on boxes while the sticks are generally limited to 600 MHz or so.

The Sticks

Among the competition were the Roku Streaming Stick 3500R, Google's Chromecast and quite a lot of EZCast based dongles. Ematic's Mediabeam, Merlin-Digital's Screencast and Cubetek Air Streamer were in the contention. We'd have liked to include Amazon's Fire TV Stick as well since it is quite capable and would have easily given most of the competition a run for their money but we couldn't manage to get our hands on one. Needless to say, we will try to get one just to see how well they stack up against the big guys and the not so popular ones.

Roku's 3500R is an unusual entry in this test since the service is not available in India but that wouldn't stop anyone from investing in a VPN service to unlock its full potential. The sheer amount of content at your disposal is something that only the Amazon Fire Stick can compete with. However, these are all paid services but you do have some good free services like Pandora, PBS, TED, Crackle etc. The device is accompanied by a physical remote that runs on RF so there is no need to point it directly at the device. If that isn't enough, there are apps available across all platforms which not only allow you to stream content but also act as remotes themselves. So there is the element of flexibility. A little problem with the apps are that they're region restricted so you'll have to use a service like Evozi to get the APK file.

Chromecast was one of the first on the scene and is in need of

HOW WE TESTED

Given the power available to these streaming receivers, it was absolutely unfair to compare a streaming stick to a streaming box. There was no chance that the former could compete with the latter on performance or features which is why we broke them down into these two categories. Both were scored on mostly the same set of parameters except for when those parameters wouldn't be applicable. Here's how we set up the testing methodology.

Display - The Alienware OptX AW2310 3D made for a suitable candidate for the display which needed to support 720p and 1080p.

Wireless Router - TP-Link's Archer C7 is a fairly priced 802.11ac router and for this test we needed to ensure that the router should be dual band with a high enough bandwidth. The C7 can go all the way upto 1300 Mbps on the 5 GHz band and 450 Mbps on the 2.4 GHz band.

Mobile Devices - HTC's One M8 was used to test stuttering during Miracast and to playback via Wi-Fi direct. A device with a powerful SoC was needed since most of the processing during Miracast happens at the transmitting device and not at the receiver.

LG/Google Nexus 4 was used in cases where the HTC would have failed to connect.

Apple iPod Touch 5th Gen was used to test out Airplay.

Media Server - Plex is a decent media server which works across all devices regardless of what platform it may be installed on. Our test Plex server was hosted on a laptop which was connected to the router via Ethernet to eliminate any bandwidth sharing which might have impeded the wireless performance.

The devices were restricted to the TP-Link Archer C7 to ensure bandwidth was always available. The channels on the router were selected to reduce interference with other Wi-Fi devices in the area. Most of the transmission was via the router except when media playback had to occur via a Plex media server.

We tested the different media formats on each system along with the distance from which each device could effortlessly receive the stream. We looked at the app on mobile platforms and the functionalities that they had and we also scored the devices based on how easy it was to navigate through the interface.

Now most of these devices do not support all the features but can easily do so by use of an app on either the device or on your media server. We included these in the features since these were easy to implement. Any workaround that would have required a steep learning curve was not used. The tests not only involved feature comparisons but also compatibility across media formats.

a much needed upgrade. There are quite a lot of apps on the Play Store and the iOS App Store which support the Chromecast so if you wish to watch Netflix then you need to download the

app and then cast from your phone to the Chromecast. So for each new service you wish to watch you'll need to download a new app which makes the experience a little painful

since a single point of access is preferred over this approach. Then if you are casting from the Chrome desktop browser, you are limited to 720p which was low even back when it launched 18 months ago.

EZCast is a Chinese jack-of-all-trades brand which has incorporated every single streaming technology into their dongle so it's a pretty good deal if you want a simple streaming device. There is the obvious limitation that they don't have any tie-ups with studios to bring exclusive content to their devices. Moreover, since the software isn't open-source there lies a major disadvantage that custom firmware does not exist. Also, there are no add-on channels for the EZCast dongles that the Roku and the Chrome-



Neo X7 is the slightly lower powered cousin of the X8-H Plus